

ST JOSEPH'S BOYS' HIGH SCHOOL, BENGALURU
MIDTERM ASSESSMENT 2026
BIOLOGY PAPER 1(Theory)

Std: 12

Date: 25.09.2026

Max. Marks: 70

Time: 3 Hours

Instructions to Candidates

1. You are allowed **an additional fifteen minutes** for only reading the question paper.
 2. You must **NOT** start writing during reading time.
 3. This question paper has **11 printed pages**.
 4. It has **eighteen questions** in all. Answer **all** questions.
 5. There are four sections in the paper: A, B, C, and D. **Internal choices** have been provided in **two questions**, each in **Sections B, C, and D**.
 6. **Section A** consists of **one question** consisting of eighteen subparts. Each subpart carries **one mark**.
 7. **Section B** consists of **seven questions** each carrying **two marks**.
 8. **Section C** consists of **seven questions** each carrying **three marks**.
 9. **Section D** consists of **three questions** each carrying **five marks**.
 10. **Diagrams should be drawn** wherever necessary using a **pencil** only.
 11. The intended marks for questions are given in [].
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SECTION A – 20 MARKS

Question 1

(i) Person 'X' experiences small, red or grey painless growths around the genitals, anus and upper thigh and observes itching and bleeding. The doctor tells X that it's a viral infection. [1]

Name the disease that Person X is infected and the virus that causes this disease.

(ii) Which one of the following reflects the correct type of sex-determination in Grasshopper and cockroaches? [1]

- (a) XX – XY Type
- (b) XX – XO Type
- (c) ZO – ZZ Type
- (d) ZW – ZZ Type

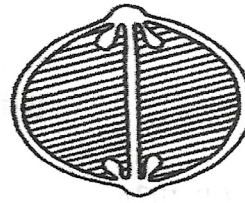
(iii) Two butterfly species are competing for the nectar from same flower in a garden. To survive and coexist together, they may avoid competition in the same garden by: [1]

- (a) predating on each other
- (b) feeding at the same time
- (c) choosing different foraging patterns
- (d) feeding different parts of the same resource

(iv) Observe the relation between the first two words and then fill in the fourth word. [1]

Lymphocytes : Antibody :: Mast cells : _____

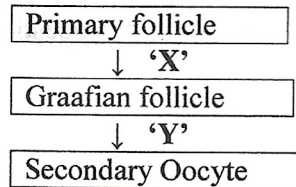
- (v) In the given figure of a fruit, label the part which is protective in function and that which is responsible for producing new plants.



[1]

- (vi) Fill in the hormonal factor 'X' and 'Y' responsible for the ovarian changes during menstrual cycle events shown below.

[1]



- (vii) Given below are statements marked Assertion and Reason. Read both the statements carefully and choose the correct option.

[1]

Assertion: Opioids are used as drugs of abuse.

Reason: Opioids slow down metabolism and produce hallucinations.

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Both Assertion and Reason are false.

- (viii) **Assertion:** Anemophilous flowers produce light and non-sticky pollen grains.

[1]

Reason: Light and non-sticky pollen grains can be easily transported over vast distances by wind currents.

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Both Assertion and Reason are false.

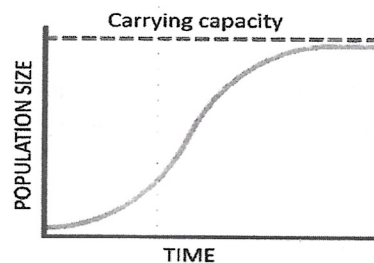
- (ix) What is the role of enzyme aminoacyl tRNA synthetase in protein translation?

[1]

- (x) A patient was advised to take probiotic curd along with antibiotics. Give reasons.

[1]

- (xi) Name the type of population growth curve shown in the graph and write the equation used to measure it.



[1]

- (xii) A farmer has been advised to sow soyabean seeds inoculated with a bacterial culture. Name the bacterium and explain how will it be useful to the crop.

[1]

(xiii) Oral administration of small doses of either progesterone or progesterone-estrogen combination is used by females as one of the contraceptive methods. Explain the mechanism involved and name this method. [1]

(xiv) Elephants have evolved longer trunks enabling them to reach leaves higher up in trees. Name the type of natural selection involved. [1]

(xv) After occurrence of heavy rain, large number of moth and other insects were found near the light source at night. People thought they have arisen from dew and mud under the influence of heat and moisture. Do you agree with this statement? Name the theory of origin of life associated with this. [1]

(xvi) Answer the following:

(a) Expand the abbreviation VNTRs. [1]

(b) Name the scientist who gave Chromosomal theory of inheritance. [1]

(xvii) State the underlying principle of IVF. [1]

(xviii) Give reasons for each of the following statements:

(a) Mendel selected pea plants for his experiments. [1]

(b) Presence of mitochondria in the middle piece of sperm. [1]

SECTION B – 14 MARKS

Question 2 [2]

Write the difference between the tender coconut water and the thick white kernel of a mature coconut and their ploidy.

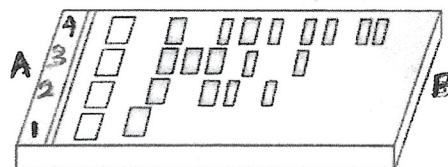
Question 3 [2]

The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollinated by a species of bee. What type of population interaction is this, and what happens to the interaction if the bee species goes extinct locally?

Question 4 [2]

(i) Identify the anode end in the diagram given alongside. Why have DNA fragments in the well '4' moved farther away?

(ii) How are the fragments visualized?

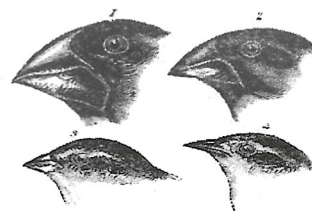


Question 5 [2]

The given figure is of Darwin's finches.

(i) Mention the specific geographical region and the phenomenon that has resulted in the evolution of these birds.

(ii) Explain the existence of different varieties of finches.

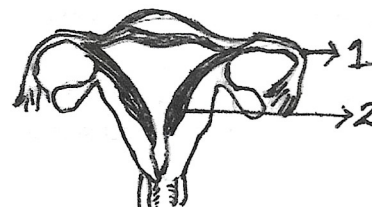


Question 6

[2]

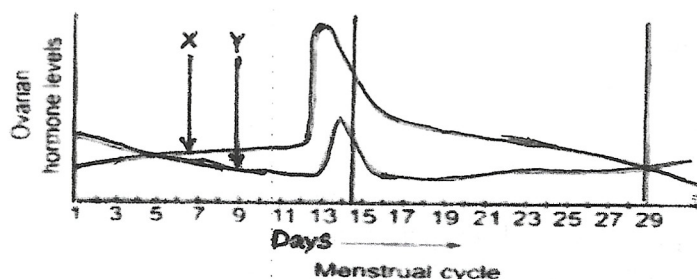
(i) Study the diagram given below and answer the questions that follow:

- (a) Identify the parts labelled 1 and 2 stating the role of Part labelled '2'.
 (b) Name the type of sex chromosome of the gametes produced by the ovaries.



OR

(ii) The levels of pituitary hormones during the different days of menstrual cycle of a human female are shown in the graph given below.



- (a) Name the hormones labelled X and Y in the graph.
 (b) What is the effect of peak levels of hormone X on the ovarian events? Explain.

Question 7

[2]

The table given below shows some mRNA codons and their associated amino acids. Analyze it and answer the following questions:

- (i) What will be DNA base sequence that would code for the following amino acid sequences:

Met – Gln – Gln – Phe – Ala – Try – Ile

- (ii) What will be tRNA anticodons for the following mRNA codons?

AUG CAU AUA UUU CAA CAU

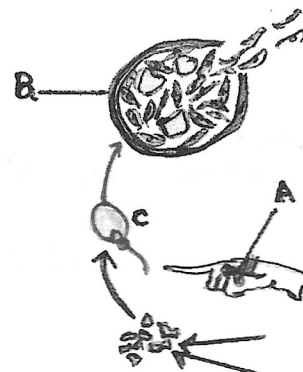
Codons	Amino acid
UUU	Phenyl alanine (Phe)
CAA	Glutamine (Gln)
UAC	Tyrosine (Try)
GCG	Alanine (Ala)
AUG	Methionine (Met)
AUA	Isoleucine (Ile)
CAU	Histidine (His)

Question 8

[2]

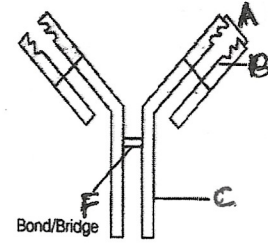
(i) Study the diagram given alongside and answer the questions given below:

- (a) Name the infective stage of the malarial parasite that enters human blood when the mosquito bites.
 (b) Name the event C and the organ where this event occurs.
 (c) Plasmodium completes its life-cycle in two hosts – man and mosquito. Name the host in which it passes sexual phase.



OR

- (ii) Study the molecule shown in the diagram given alongside.
 (a) Label the parts A, B, and C.
 (b) What would happen if the Bridge F breaks?



SECTION C – 21 MARKS

Question 9

[3]

- (i) Draw a neat, labelled diagram of an Embryo sac in flowering plants.

OR

- (ii) Draw a neat, labelled diagram of T.S. of a seminiferous tubule.

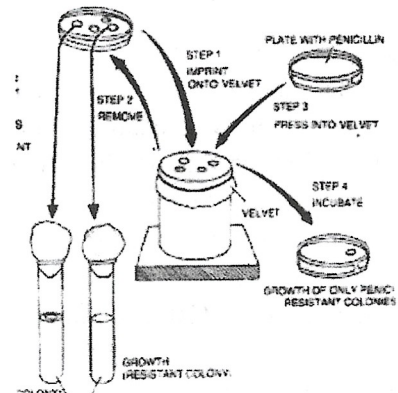
Question 10

[3]

- (i) A mutant strain of *E. coli* where the operator gene (O) is mutated (O^c) such that the repressor protein can no longer bind to it.
 (a) Draw a simplified linear diagram of the lac operon showing genes i, z, y, a, promoter, and the mutated operator.
 (b) Predict the expression of structural genes (z, y, a) in the absence of an inducer (lactose). Will the enzymes β -galactosidase, permease, and transacetylase be produced? Give a reason.

OR

- (ii) The given diagram shows replica plating experiment.
 (a) Name the scientists and the aim of conducting the experiment.
 (b) Explain the procedure followed to conduct the experiment.

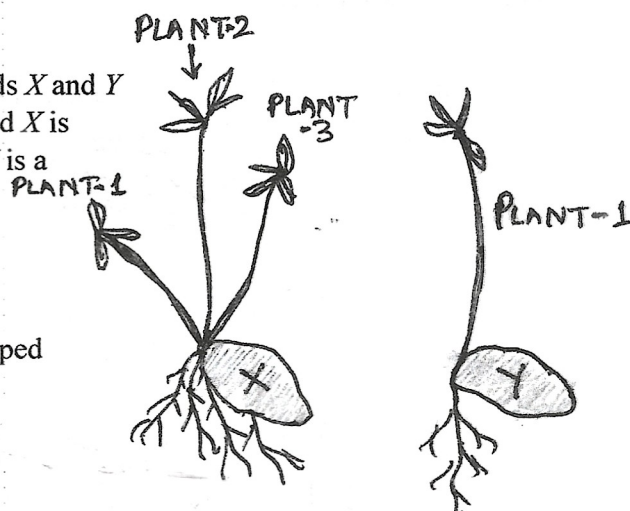


Question 11

[3]

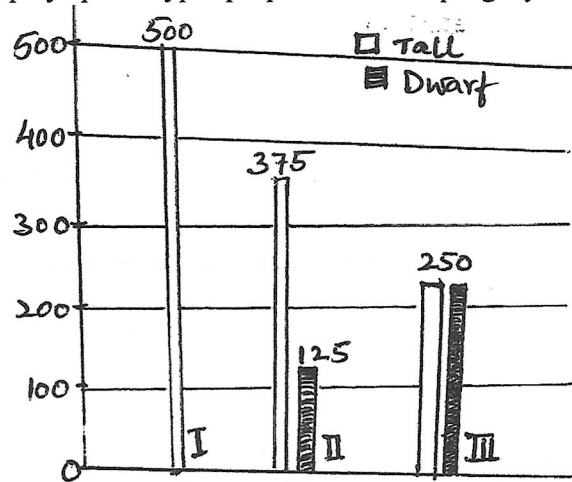
The image shows two germinated seeds X and Y which belong to the same species. Seed X is produced by apomixis whereas seed Y is a product of sexual reproduction.

- (a) Write the number of embryo(s), embryo sac(s) and ovules in the ovary of seed X.
 (b) What advantage will plants developed from seed Y have over seed X?
 (c) Mention one similarity between parthenocarpy and apomixis.



Question 12**[3]**

A student performed some crosses in plants and represented the result in the form of bar graphs as shown below. Each graph displays phenotypic proportion of the progeny. Study the graphs and answer the questions:



- (i) What can you infer about the genotype of parents in crosses I and II?
- (ii) Looking at bar graph of cross III, identify the type of cross performed and its importance in genetics.
- (c) What conclusion can you draw from the results of bar graphs of crosses I and II? Name the genetic principle illustrated.

Question 13**[3]**

Population size in nature keeps changing with time depending on various factors including food availability, predation, and adverse weather.

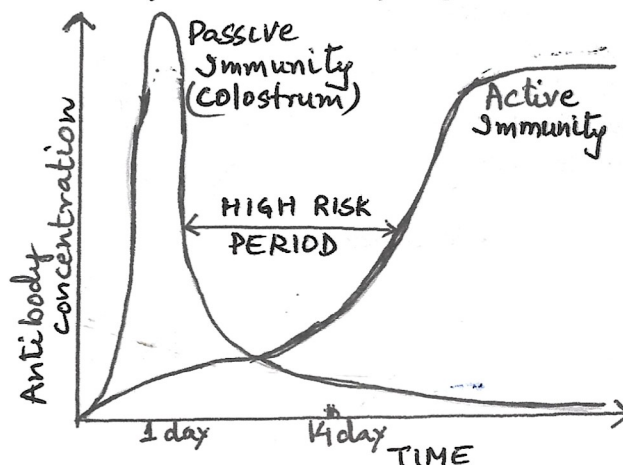
- (a) The Monarch butterfly is highly distasteful to its predator because of a special chemical present in its body. How does the butterfly acquire this chemical?
- (b) Name two methods to measure the population density of tigers.
- (c) In a pond there were 30 hydrilla plants. Through reproduction 10 new hydrilla plants were added in a year. Calculate the birth rate of the population.

Question 14**[3]**

Explain the process of making *lm*mRNA into fully functional mRNA in eukaryotes. Where does this process occur in the cell?

Question 15**[3]**

The graph below shows the antibody concentration in young calves. Study the graph and the questions that follow:



- (i) What is the difference between passive and active immunity in this case?
- (ii) What kind of immunity will be observed when a vaccine is administered to the calf and why?
- (iii) How do cellular barriers and cytokine barriers provide innate immunity in humans?

SECTION D – 15 MARKS

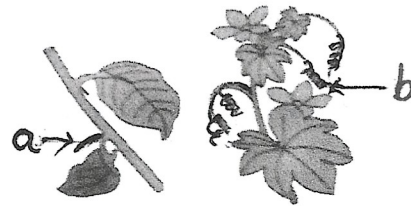
Question 16

[5]

(i) The ability to roll tongue is determined by a dominant allele. In a sample of 700 people, 210 were unable to roll their tongues. Using Hardy-Weinberg equation, calculate how many people in the sample were homozygous dominant and heterozygous for tongue rolling?

(ii) Identify the homologous organs in the diagram giving reasons. Give an example of the same in animals.

(iii) What are the two striking similarities observed when different stages of embryos of different classes of vertebrates are studied?



Question 17

[5]

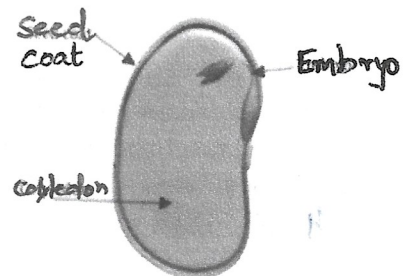
(i) (a) Differentiate between the following based on the parameter specified below:

- (1) Perisperm and Pericarp based on the part they represent.
- (2) Oogenesis and Ovulation based on the hormone involved.
- (b) How is polyspermy prevented in humans?
- (c) What changes occur in the uterus during the secretory phase?

OR

(ii) Observe the given diagram carefully and answer the following questions:

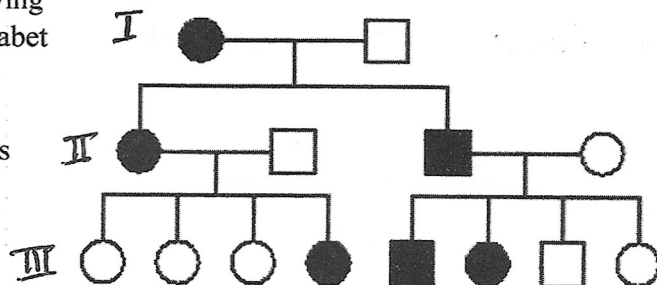
- (a) (1) Name the parts of flower which develop into seed and seed coat.
- (2) Can a fruit be formed without fertilization? If yes, what is it called and what is its major drawback?
- (3) Classify the following into albuminous and exalbuminous seeds: Groundnut, pea, maize, castor



(b) The following pedigree chart represents the pattern of inheritance in a family.

(1) Identify the type of inheritance giving reasons for your answer. Use the alphabet A/a to represent the alleles.

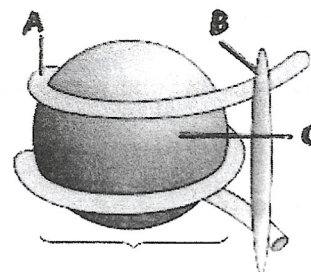
(2) What will be the genotype of the generation I parents? Explain the cross using a Punnett square.



Question 18

[5]

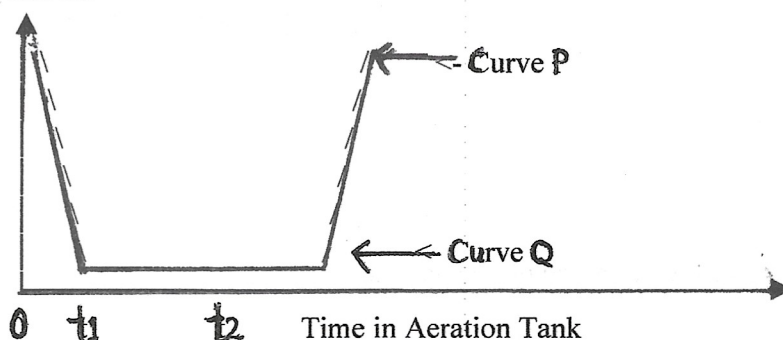
(a) The diagram alongside is of a nucleosome present in the chromatin fibre of chromosome.



- (1) Label the parts A, B, and C.
- (2) Name the histone molecules found in the core.

(b) A biology student monitored the relationship between **microbial biomass (flocs)** and **BOD** in the primary effluent after it was transferred to an aeration tank of a Sewage Treatment Plant (STP). The observations were plotted below:

Amount



- (1) Label Curve P and Curve Q based on the working of an aeration tank.
- (2) What are "flocs" and what is their role in changing Curve Q from time 0 to time t_1 ?
- (3) At what point (t_1 or t_2) should the effluent be shifted to the settling tank? Justify your choice.

OR

(a) (1) *E. coli* cells grown in a heavy isotope N^{15} medium for many generations are transferred to a light isotope N^{14} medium and allowed to replicate for 3 complete generations. What is the ratio of entirely light N^{14} - N^{14} DNA molecules to hybrid N^{14} - N^{15} DNA molecules in the resulting population? Explain it with a labelled diagram.

(2) An experimental drug selectively inhibits DNA ligase activity in eukaryotic cells during the S phase. When analysing the genomic DNA of these treated cells through gel electrophoresis, what specific molecular pattern will be observed? Explain.

(b) State two goals and methodologies of Human Genome Project.
